

Millimetre-scale distribution of organic matter composition at intact biopore and crack surfaces

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Summary

Structured subsoil horizons are characterized by biopores and shrinkage cracks, which may serve as preferential flow paths. The surfaces of cracks and biopores may be coated by clay-organic material. The spatially-distributed organic matter (OM) composition at such structural surfaces was studied at the millimetre scale using diffuse reflectance infrared Fourier transform (DRIFT) spectroscopy in the mid-infrared range (MIR). Intact biopores such as earthworm burrows and root channels, and crack surfaces of nine subsoil horizons were analysed. The samples were from arable and forest Luvisols, one Regosol, one Stagnosol and Cambisols developed from loess, till, mudstone and limestone. For better comparison between soils, the DRIFT signal intensities were corrected for the particle-size effects. The OM was characterized by the ratio between alkyl- (C–H) and carbonyl (C=O) functional groups (C–H/C=O), which represent an index of the potential wettability (PWI) of the OM. The PWI was larger for biopores than for crack surfaces and the soil matrix, indicating a smaller potential wettability of OM at biopore surfaces. The millimetre-scale spatial variability of OM was especially large for the surfaces of root channels. Samples from till-derived Luvisols had smaller PWI (with greater potential wettability than surfaces from loess-derived Luvisols) than other soil types. The mean PWI of the arable Luvisol crack surfaces was less than that of the forest Luvisol samples. The results suggest that the spatial distribution of OM properties at intact structural surfaces may be important for describing sorption and mass transfer processes during preferential flow.

Introduction

Soil organic matter (SOM) is composed of a number of diverse organic matter (OM) fractions such as particulate organic matter (POM), organo-mineral complexes, humic acids and colloidal or dissolved organic matter, each consisting of various components. The vertical spatial distribution of OM in the bulk soil is well known; in addition, a non-uniform spatial distribution of OM in addition frequently exists at the small scale (mm to cm), especially in structured soils (Chabbi *et al.*, 2009). The burrowing activity of soil animals and the growth of plant roots can lead to larger OM contents around burrows and roots than in other soil regions (Lee, 1985). Aggregate and fissure coatings are a pedologic characteristic especially for clay-illuvial B horizons of a Luvisol (Lin, 2010); other types of macropore wall coatings have been reported for the drilosphere around an earthworm burrow (Don *et al.*, 2008).

When considering preferential flow (PF), biopores, cracks and fissures can serve as pathways for the downward percolation of infiltrated water during storms. Also, the preferential transport of

solutes, dispersed colloids and particles is largely restricted to this vertically-continuous pore system (Köhne *et al.*, 2009). The interaction between percolating solution and soil takes place at the surfaces of PF paths, which provide an effective area for mass exchange between the PF pore domain and the soil matrix (Gerke, 2012) and for sorption and exchange processes for preferentially transported chemicals (Lin, 2010). This interface has a relatively large ‘reactivity’ with respect to OM decomposition (Bundt *et al.*, 2001) and may accelerate the sorption and decomposition of pesticides (Fahrenhorst *et al.*, 2000).

The complex mixture of organic components in SOM can be characterized by functional groups such as peptides, polycarbohydrates or fatty acids. The OM composition determines the physicochemical properties of OM, such as cation exchange capacity (Kaiser *et al.*, 2008) and wettability (Capriel *et al.*, 1995).

The SOM functional groups can be characterized by Fourier transform infrared (FTIR) spectroscopy (Demyan *et al.*, 2012) in the mid-infrared range (MIR; wave lengths between 2.5 and 25 µm). As a specific FTIR technique, diffuse reflectance infrared Fourier transform (DRIFT) spectroscopy can be applied to undiluted samples. In contrast to nuclear magnetic resonance (NMR) spectroscopy, DRIFT spectroscopy can be applied to intact

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sample surfaces of different scales. To distinguish between the absorption bands of aromatic and aliphatic C–H groups of SOM, a deconvolution of specific spectral regions (Kauppinen *et al.*, 1981) can be used. The heterogeneity of OM composition was observed at scales of 20–250 µm in intact soil microaggregates (Lehmann *et al.*, 2007) and at the millimetre scale along single transects in intact structures such as earthworm burrows and aggregates (Ellerbrock *et al.*, 2009; Leue *et al.*, 2010b). Following optical laws, the particle size of soil samples affects the signal intensity of DRIFT spectra: an increasing particle size causes larger proportions of primary reflection and thus smaller signal intensities of diffuse reflection (Kortüm *et al.*, 1963; Leue *et al.*, 2010a). When comparing DRIFT spectra measured on differently-textured samples taken from different soils for example, a particle size correction will be necessary.

Conceptually, the interface between the soil matrix and the PF path (Figure 1) may show a millimetre scale spatial heterogeneity of SOM composition among other components. The local OM composition determines the wettability and the sorption properties of the PF path surfaces as well as the diffusive mass transfer of sorbed chemicals towards the soil matrix. The OM composition at intact structural surfaces has only recently been analysed in more detail (Ellerbrock *et al.*, 2009; Leue *et al.*, 2010b). These studies suggested that three main types of soil structural features could be distinguished: (i) earthworm burrows, (ii) root channels and (iii) crack surfaces, which differ with respect to the spatial distribution

of OM composition of the coating. Of course, the OM composition will depend also on the soil type and the parent material that provided mineral components for organo-mineral complexes. Eventually, the land-use type will affect both the content and the composition of the OM through biomass input (Jandl *et al.*, 2012).

The objective of our study was to characterize the millimetre-scale spatial distribution of the OM composition at intact structural surfaces of subsoil horizons for different soil types. Moreover, the influence of parent material, soil type and land use on the OM was analysed. For this purpose it was necessary to identify intact structural surfaces of structured subsoil horizons, to distinguish different surface types for these samples (earthworm burrows, root channels and cracks), to prepare the intact structures from soil blocks and to characterize the distribution of the OM composition at the structures in a non-destructive way by means of DRIFT spectroscopy in the MIR.

Materials and methods

Soils and sampling

Soil samples were collected from three Haplic Luvisols (haLV), one Albic Luvisol (abLV), one Haplic Regosol (haRG), one Vertic Stagnosol (vrST), one Vertic Cambisol (vrCM) and one Haplic Cambisol (haCM) (Table 1). These sites, located in north-eastern Germany (Holzendorf, Rowa and Gramzow), south-western Germany (Simmringen, Aichelberg, Ammerbruch and Nellingen) and

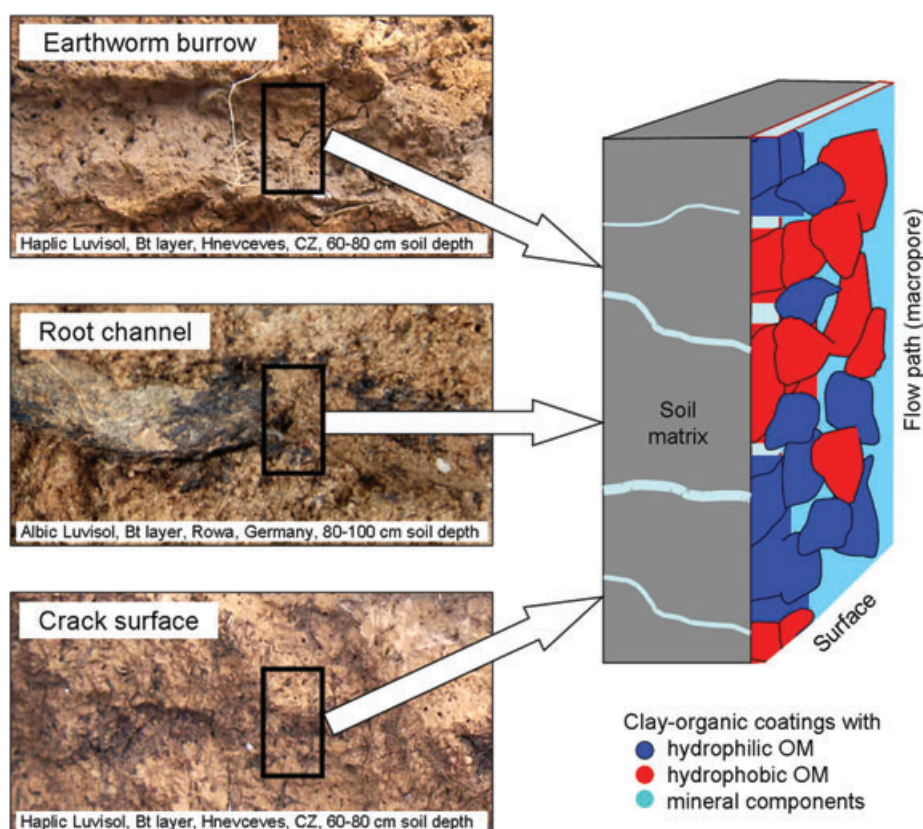


Figure 1 Photographs (left) of intact structural surfaces from subsoil horizons for an earthworm burrow (top), root channel (centre) and a crack surface (bottom) and illustration of the concept of spatially distributed organic matter (OM) composition at the surface of these structures consisting of clay-organic coatings. The spatial distribution of the hydrophilic and hydrophobic functional groups of the OM is assumed to determine the millimetre-scale spatial distribution of flow path surface properties such as the potential wettability.

Table 1 Soil classification, geographical position, altitude (above sea level) and climatic conditions (annual mean precipitation, P; mean annual temperature, T) for the study sites

Site name	Soil classification WRB (2006)	Longitude / °E	Latitude / °N	Altitude / m	P / mm year ⁻¹	T / °C
Simmringen	Haplic Luvisol (haLV)	09°52'57''	49°34'47''	338	577	9.4
Hnevceves	Haplic Luvisol (haLV)	15°43'03''	50°18'47''	273	618	8.5
Holzendorf	Haplic Luvisol (haLV)	13°47'11''	53°22'45''	48	501	8.7
Rowa	Albic Luvisol (abLV)	13°16'20''	53°29'47''	90	532	7.9
Gramzow	Haplic Regosol (haRG)	14°03'02''	53°13'05''	75	532	8.3
Aichelberg	Vertic Stagnosol (vrST)	09°33'14''	48°38'37''	383	1024	8.8
Ammerbuch	Vertic Cambisol (vrCM)	08°56'13''	48°33'59''	420	795	8.3
Nellingen	Haplic Cambisol (haCM)	09°46'18''	48°33'16''	710	1069	6.8

Table 2 Sites and parent material, land use, soil horizons, sampling depths, mean values of the soil pH (0.01 M CaCl₂), the texture (wet sieving and sedimentation) and soil organic carbon (SOC; dry combustion, Vario Elementar, Hanau, Germany) contents (three replicates for all measurements)

Site name	Soil	Parent material	Land use	Horizon	Depth / cm	pH	Clay / %	Silt / %	Sand / %	SOC / %
Simmringen ^a	haLV1(a)	Loess	Arable	Bth	60–80	7.2	35.4	63.2	1.4	0.52
Simmringen ^a	haLV1(f)	Loess	Forest	Bht	60–70	4.6	34.3	63.0	2.7	0.43
Hnevceves	haLV2	Loess	Arable	Bt	60–80	4.6	34.6	56.2	9.2	0.44
Holzendorf	haLV3	Glacial till	Arable	Bt	40–60	6.4	16.5	35.0	48.5 ^b	0.31
Rowa ^a	abLV	Glacial till	Forest	Bt	80–100	4.7	16.8	24.4	58.8	0.24
Gramzow ^a	haRG	Glacial till	Forest	Ah-Cv	50–60	7.4	17.1	41.6	41.3	2.79
Aichelberg ^a	vrST	Mudstone	Forest	Sw	40–60	7.6	60.5	33.5	6.0	2.92
Ammerbuch ^a	vrCM	Mudstone	Arable	P	25–40	6.7	59.9	29.7	10.4	3.24
Nellingen ^a	haCM	Limestone	Forest	Tv	50–60	4.9	73.7	24.8	1.5	1.01

^aKaiser *et al.* (2010).

^bBecause of the small-scale heterogeneity of the particle size distribution at the Holzendorf site, the mean sand content of the mixed samples including all soil material was about 48.5% while the sand content of the mixed 'matrix' sample without clay-organic coating, used for the particle size correction, was 91.7% (clay + silt = 8.3%).

in the northern Czech Republic (Hnevceves) cover a wide range of soils that are different in parent material and clay content, and which developed under varying climatic conditions and land use (Table 1). The parent material was loess (Simmringen and Hnevceves), mudstone (Aichelberg and Ammerbuch), limestone (Nellingen) and glacial till (Holzendorf, Rowa and Gramzow) (Table 2). The soil pits at the arable site haLV1(a) and the forest site haLV1(f) were located about 200 m apart. All soils had structured subsoil horizons with potential PF paths such as cracks, fissures, earthworm burrows and root channels; the latter occurring mainly in the forest soils. The subsoil structures were relatively stable and characterized by distinctive differences in physicochemical properties between matrix coatings of cracks and biopores (for the Loess soil at Hnevceves see Kodesova *et al.*, 2011).

At the soil pits, sampling was restricted to the subsoil horizons from 25–80 (top) to 40–100 cm (bottom) depth (Table 2). Fifteen-centimetre-high soil blocks of approximately 20 × 30-cm edge-length were cut or broken out of the subsoil horizons with a spade. Smaller subsamples or clods of about 5–10-cm edge-length were manually separated from these larger blocks. The separation was carried out along the cracks that were visible at the clod surfaces (Figure 2). Regions with clay-organic coatings, earthworm burrows and root channels were identified by visual

inspection and were separated by manually scraping material off the surrounding clod. The samples were wrapped up in tinfoil to stabilize the sides (Ellerbrock *et al.*, 2009) and dried over silica gel in a desiccator. Additionally, mixed samples were taken from the subsoil horizons for determining the pH and the clay and SOC contents (Table 2).

DRIFT spectroscopy data collection and pre-processing

DRIFT spectroscopy in the MIR (wave lengths of 2.5–25 µm) was carried out in 1-mm steps along transects within the intact surfaces of earthworm burrows and root channels as well as perpendicular to the biopores, including parts of the crack surfaces on both sides of the biopore (Figure 2). To avoid micro-topography effects on DRIFT spectra along the rough intact surfaces (Leue *et al.*, 2011), the incident IR beam (diameter 1.5 mm) was focused at each measurement point with respect to the surface elevation by using an adjustable screw at the XY-positioning table on this DRIFT device (Resultec, Illerkirchberg, Germany). Mixed samples, representing matrix properties, were taken from structured soil without visible coatings and were carefully crushed in an agate mortar without particle destruction. The samples were poured into 4-mm-tall cups with a diameter of 11 mm. These cups

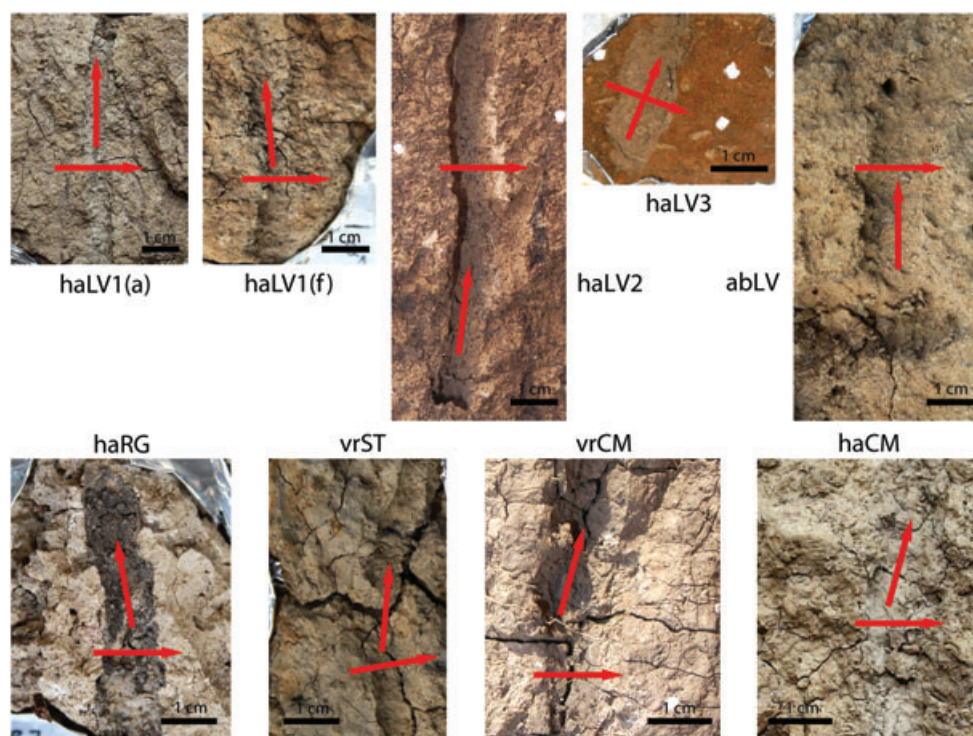


Figure 2 Intact structural surfaces of five Luvisol (haHL, abLV), one Regosol (haRG), one Stagnosol (vrST) and two Cambisol (vrCM, haCM) subsoil horizons, from parent material, loess (haLV1, haLV2), glacial till (haLV3, abLV, haRG), mudstone (vrST, vrCM) and limestone (haCM), and for arable (a) and forest (f) soils. The red arrows depict the location and direction of transects along which DRIFT spectroscopy was carried out in 1-mm steps.

were then placed on the same aluminium plate as used before for the intact samples and positioned underneath the IR beam using the XY-positioning table. Three replicates of every sample were measured with DRIFT spectroscopy without dilution.

The DRIFT spectra were recorded by 16 co-added scans between wave numbers (WN) 4000 and 400 cm^{-1} at a resolution of 4 cm^{-1} , using a gold target (99% Infragold®, Labsphere, North Sutton, NH, USA) for the background spectra measurements. The spectra were converted to Kubelka-Munk (KM) units (Kubelka, 1948), smoothed (boxcar moving average algorithm, factor 25), and corrected for baseline shifts using WIN-IR Pro 3.4 (Digilab, Randolph, MA, USA). Additionally a deconvolution was carried out for the spectral region between WN 3100 and 2700 cm^{-1} using an exponential algorithm with a K factor of 2.0, a half width of 60 and a Bessel smoothing function (Kauppinen *et al.*, 1981).

Particle size correction

The spectra were corrected for the effect of texture (Table 2) on the reflectance signal (the primary reflection or scattering increases with particle size). We considered the KM transformation, $f(Re)$, of the DRIFT reflectance spectra (Kortüm *et al.*, 1963) as:

$$f(Re) \equiv \frac{(1 - Re)^2}{2Re} = \frac{K}{S}, \quad (1)$$

where Re is the diffuse reflectance of the solids at the outermost sample surface layer of approximately 4–8 μm thick, K is the absorption module and S , the scattering coefficient, is determined by the primary reflection of the IR beam at the sample surface. In the standard DRIFT procedure, the value of S would usually remain constant because the reflection from the sample surfaces is experimentally made uniform by grinding and dilution. Here, the particle size effect on the diffusive reflection at the intact sample surfaces was corrected by first analysing particle size effects in reflectance mode (not KM transformed spectra; Figure 3a) using the signal intensities of two bands at WN 1873 and WN 1990 cm^{-1} . These bands are a combination of overtones caused by symmetric and asymmetric lattice vibrations of SiO_2 groups within the silt and sand particles (Benesi & Jones, 1959) called ‘bulk mode (BM) bands’ (Péré *et al.*, 2001; Leue *et al.*, 2010a). In undiluted soil samples such BM bands are ubiquitous and increase with increasing particle size (Leue *et al.*, 2010a). In contrast to other spectral regions, the BM bands are not affected by absorption bands that are specific for certain functional groups of clay minerals, water or organic compounds.

The calibration using the summed BM band signal intensities at WN 1873 and 1990 cm^{-1} , which increased with the sand content ($R^2 = 0.89$) of the nine mixed subsoil samples, yielded the linear relation of $y = -0.001x + 0.167$, where x is the sand content (%) and y is the summed BM signal intensity in reflectance mode.

Table 3 Sum of the bulk mode (BM) bands at WN 1873 and 1990 cm⁻¹ (in reflectance mode) and quotient b_i , describing the relation of the BM bands to that of vrCM

Mixed sample Soil type	Sum BM bands / (Re)	b_i / (-)
haLV1(a)	0.154	0.846
haLV1(f)	0.145	0.797
haLV2	0.156	0.857
haLV3	0.058	0.319
abLV	0.101	0.555
haRG	0.126	0.692
vrST	0.162	0.890
vrCM	0.182	1.000
haCM	0.102	0.560

A quotient, b_i , between the signal intensity of the BM bands of each of the mixed samples and the intensity of BM bands of the vrCM sample was calculated as:

$$b_i = \frac{BM_i}{BM_{vrCM}}, \quad (2)$$

where the subscript i denotes the number of soils ($i = 1, \dots, 9$). The values of b_i (Table 3) were used to normalize the reflectance spectra of the coarser textured samples to those of the Ammerbuch sample (vrCM), which shows the greatest signal intensity because of its clay content. The reflectance spectra of each matrix sample were normalized with respect to the particle size effect on the signal intensities (Figure 3b) as:

$$Re*_{i,n} = b_i * Re_{i,n}, \quad (3)$$

where subscript n denotes the wave number in the range WN 4000–400 cm⁻¹.

The normalized reflectance spectra Re^* (Figure 3b) were transformed to Kubelka-Munk units using Equation (1) (Figure 4a) and related to the original (not manipulated) KM spectra $f(Re)$

(Figure 4b) of the crushed matrix samples:

$$S_{i,n} = \frac{f(Re*_{i,n})}{f(Re_{i,n})}, \quad (4)$$

where $S_{i,n}$ is a WN-dependent and soil-specific scattering coefficient (Figure 5).

Assuming that the sand content mainly determines the scattering of a sample, $S_{i,n}$ was used as the normalized scattering module in Equation (1) to consider the particle size effect on KM spectra. The normalized spectra, $f(Re)_{corr}$, were obtained by correcting DRIFT spectra measured at intact structural surfaces, $f(Re)_{surf}$, with the empirical scattering coefficients (Equation (4)) as:

$$f(Re)_{corr} = \frac{f(Re)_{surf}}{S_{i,n}}. \quad (5)$$

OM characterization by DRIFT spectra analysis

In the particle size-corrected DRIFT spectrum, $f(Re)_{corr}$ (Equation (5)), the C–H stretching vibrations of alkyl groups were identified in the spectral regions between WN 2948 and 2920 cm⁻¹ (C–H_a, asymmetric stretching) and between 2864 and 2849 cm⁻¹ (C–H_s, symmetric stretching; Baes & Bloom, 1989; Capriel *et al.*, 1995) using a deconvolution procedure (Kauppinen *et al.*, 1981). After the identification of the peak maxima position, band heights were measured in the original spectra (with no deconvolution) as the vertical distance from a local ‘baseline’ plotted between tangential points on absorption minima located left and right near the maxima. The added band heights (C–H_a and C–H_s) were denoted as C–H bands. Second, the heights of absorption bands in the regions from 1720 to 1700 cm⁻¹ (groups with C=O double bonds of ketones, carboxylic acids or amides; Senesi *et al.*, 2003) and from 1625 to 1600 cm⁻¹ (groups with C=O double bonds of carboxylate anions and aromatic groups; Baes & Bloom, 1989; Senesi *et al.*, 2003) were measured as height from the total baseline, added, and denoted as C=O bands. The alternative method of analysing the spectral absorption band intensities by

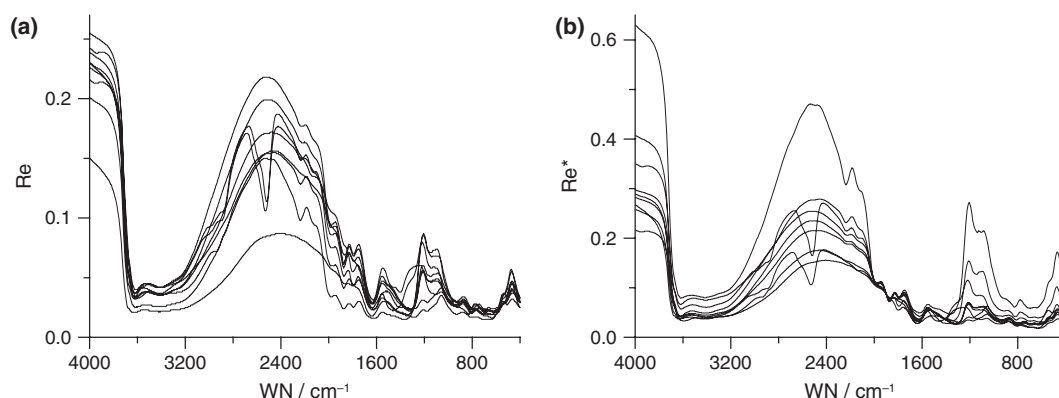


Figure 3 DRIFT spectra (in reflectance mode) of mixed and crushed subsoil samples. (a) Original spectra (Re); (b) spectra normalized (Re^*) to the BM signal intensity of vrCM.

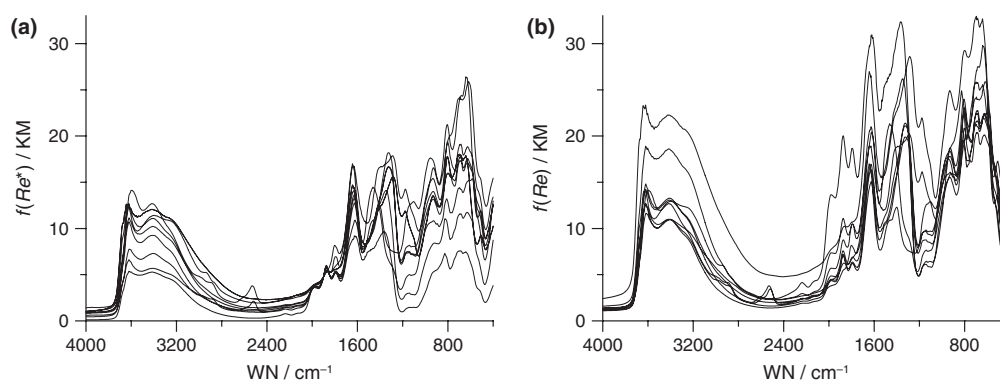


Figure 4 DRIFT spectra (transformed to Kubelka-Munk mode) of mixed and crushed subsoil samples. (a) KM spectra of the normalized reflectance spectra (Re^*); (b) original spectra.

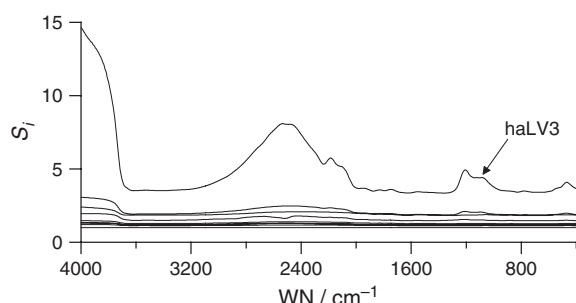


Figure 5 Scattering module S_i specific for the nine subsoil samples.

evaluating the area of the absorption bands (Demyan *et al.*, 2012) was not used here, in order to preserve comparability with earlier data (Ellerbrock *et al.*, 2009; Leue *et al.*, 2010b). For every DRIFT spectrum, the cumulative peak heights of bands C–H were related to those of the bands C=O. This C–H : C=O ratio can be used as a measure for the potential wettability of the SOM (Ellerbrock *et al.*, 2005) and is denoted as potential wettability index (PWI) in the present work. A larger value of the PWI indicates that the OM has relatively more C–H in comparison to C=O functional groups.

Statistical analysis

Statistical analyses were carried out using ORIGIN Pro software (Origin Lab, Northampton, USA). The frequency distributions of the PWI values of each sample were tested for normality by the goodness-of-fit test. If necessary, PWI data were transformed logarithmically and treated as log-normal distributions. The differences between PWI data were tested at a significance level of 0.05 by the Student's *t*-test and, for non-normal distributions, by the Mann–Whitney *U*-test (Tables 4, 5).

Results

DRIFT spectra analysis

In the MIR-DRIFT spectra of the intact structural surfaces (earthworm burrows, root channels and cracks), the O–H bands

(WN 3500–3000 cm^{-1}) were broader, and the signal intensities in the regions of the BM (WN 2000–1750 cm^{-1}), the C–H (WN 3000–2800 cm^{-1}) and the C=O bands (WN 1750–1200 cm^{-1}) were larger than those of the mixed soil matrix (Figure 6, Table 6). The C–H band signal intensities were relatively small and even below the detection limit of about 0.02 KM units for five of the nine matrix samples (haLV1(a), haLV1(f), haLV2, haLV3 and abLV). Thus, the calculated PWI values (Figure 9, marked by two asterisks) were not used for any comparisons. The C–H band signal intensity was also not distinguishable from the noise of the spectral line for a few individual spectra from transects along the intact surface of abLV (one crack) and haLV3 (two cracks). However, deconvolution of the spectral region between WN 3100 and 2700 cm^{-1} (Figure 6, middle column) allowed us to distinguish between aromatic (> WN 3000 cm^{-1}) and aliphatic (< WN 3000 cm^{-1}) C–H absorption bands. The only exception was the spectrum of the mixed matrix vrCM sample, in which the C–H band could not clearly be attributed to either the aromatic or the aliphatic spectral region. In this case, the C–H band was assumed to be an aliphatic band and was denoted as the C–H_a band. The C=O bands region of the spectra of earthworm burrows showed aggregations of neighbouring single peaks and thus the individual peaks were barely visible (Figure 6, right column). For the haRG samples (pH of about 7.4) and the vrCM samples (pH of about 6.7), absorption bands of carbonate were observed in the DRIFT spectra at WN approx. 2500 cm^{-1} (Figure 6).

SOM composition of coated structural surfaces

The PWI values obtained from DRIFT measurements along individual transects (Figure 2) in earthworm burrows and one root channel (haRG) varied between about 0.01 and 0.15 (Figure 7). The spatial distribution of PWI values for transects perpendicular to biopores revealed mostly larger values for the biopores as compared with the surrounding areas (Figure 7, top row). The PWI spatial distributions appeared to be characteristic for each sample and the PWI level was mainly determined by the C–H

Table 4 Goodness-of-fit test (Gaussian distribution) for potential wettability values of the different surface types

Site	haLV1(a)	haLV1(f)	haLV2	haLV3	abLV	haRG	vrST	vrCM	haCM
Earthworm burrows	—	+	+ (log)	+	—	+ (log)	+ (log)	+	+ (log)
Root channels	No data	—	No data	+	+ (log)	+ (log)	+	No data	+
Crack surfaces	+ (log)	+	+	+ (log)	+ (log)	+	+	+	+ (log)

Level of significance is 0.05.

Log – logarithmic values.

Table 5 Potential wettability of different surface types: significance of differences of the mean values

Site	haLV1(a)	haLV1(f)	haLV2	haLV3	abLV	haRG	vrST	vrCM	haCM
Earthworm burrows compared with root channels	No data	—	No data	+ (<i>t</i> -test)	+ (M-W)	+ (<i>t</i> -test, log)	+ (<i>t</i> -test, log)	No data	+ (<i>t</i> -test, log)
Earthworm burrows compared with crack surfaces	+ (M-W)	+ (<i>t</i> -test)	+ (<i>t</i> -test, log)	+ (<i>t</i> -test, log)	+ (M-W)	+ (<i>t</i> -test, log)	+ (<i>t</i> -test, log)	+ (<i>t</i> -test)	+ (<i>t</i> -test, log)
Root channels compared with crack surfaces	No data	+ (M-W)	No data	+ (<i>t</i> -test, log)	+ (<i>t</i> -test, log)	+ (<i>t</i> -test, log)	—	No data	+ (<i>t</i> -test, log)

+ = significant difference at $P = 0.05$.

t-test – Student *t*-test; M-W – Mann–Whitney test; log – logarithmic values.

band values along the transects, as shown for one sample from haLV1(a) (Figure 8).

The main differences in PWI values (median and standard deviation) were found between Luvisols and other soil types and for each soil between the three surface types (Figure 9). The mean PWI values were significantly smaller ($P = 0.05$) for intact surfaces and matrix samples of haLV1(a), haLV1(f), haLV2, haLV3 and abLV than those for the samples from haRG, vrST, vrCM and haCM (Tables 4, 5). Significant differences in the PWI levels were also found between intact surface types and the matrix sample for each soil, except for the Stagnosol (vrST) where the PWI values of mixed samples were relatively large. The mean PWI values of the earthworm burrows and root channels were mostly significantly larger than those of the crack surfaces ($P = 0.05$), except for haLV1(f) (burrows compared with root channels) and vrST (root channels compared with crack surfaces).

Luvisols developed on loess had greater PWI values of both intact structural surfaces and matrix samples (Figure 9, Table 5) than Luvisols developed on glacial till ($P = 0.05$). For the loess-derived Luvisol at Simmringen (haLV1), the mean PWI value was larger for earthworm burrows of the forest site haLV1(f) than the mean PWI of the arable site haLV1(a), while the crack surfaces showed little difference in PWI.

The millimetre-scale site-specific spatial distributions were also reflected by the cumulative frequency distributions of the individual PWI values (Figure 9). Frequency distributions were similarly S-shaped for the earthworm burrows (EB in Figure 10) because we observed small numbers of extremely large and small PWI values. For the root channels (RC), distribution was more diverse. For the crack surfaces (CS), the shape of the cumulative curves reflected the generally smaller PWI values that

were characteristic for the non-coated soil matrix and differed depending on the parent material. The cumulative frequency distributions were similar for the three types of intact structural surfaces of samples from the Stagnosol forest soil (vrST) and the Cambisol arable soil (vrCM), and for the earthworm burrows and root channels of the Albic Luvisol forest soil (abLV).

Discussion

DRIFT measurements at intact structural surfaces

When comparing DRIFT spectra obtained at intact structural surfaces and spectra obtained by standardized DRIFT measurements using mixed matrix samples (Figures 6), the effect of the clay-organic coating on spectral intensities can be studied. The small or non-detectable C–H signal intensities in the mixed matrix samples of the Luvisols (haLV and abLV) corresponded to small OC contents of $\leq 0.52\%$ (Table 2). Negligibly small C–H signals could be assumed to be indicative of crack surfaces that were not coated by organic material (see samples of haLV3 and abLV in Figure 6), and the surface of such samples should be wettable. In contrast, for intact structural surfaces that are enriched with clay-organic material, clearly-detectable absorption bands of OM functional groups and reasonable PWI values were obtained. Differences in OM and clay distributions between the inside and the outside of soil aggregates have been attributed to earthworm activity (Don *et al.*, 2008) or root growth (Moradi *et al.*, 2012) as well as to flow and transport, and pedogenic processes such as clay migration.

In addition to the DRIFT spectroscopic approach, ^{13}C NMR spectroscopy could be an appropriate tool to distinguish between aromatic and aliphatic C–H absorption bands. However, NMR measurements at intact structural surfaces of soils could not

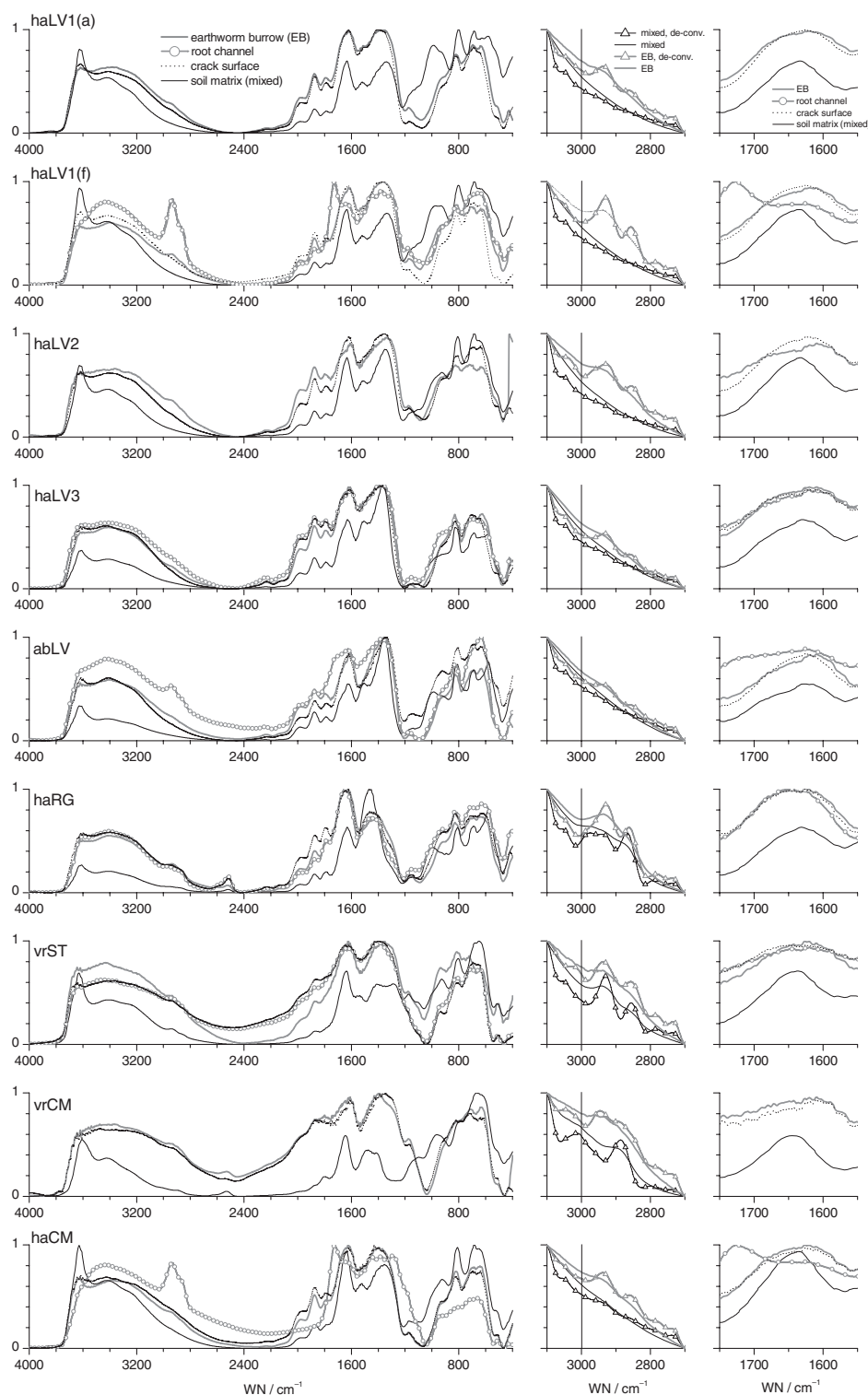


Figure 6 DRIFT spectra of earthworm burrows (EB, grey solid line), root channels (grey line with circles), crack surfaces (black dotted line) and of the carefully crushed and mixed matrix samples (black solid line) of five Luvisol (haHL, abLV), one Regosol (haRG), one Stagnosol (vrST) and two Cambisol (vrCM, haCM) subsoil horizons (left column). Image enlargements of WN 3100 to 2700 cm^{-1} (middle column) show the region of the C–H bands and of the neighbouring aromatic band between WN 3100 and 3000 cm^{-1} before (solid lines) and after deconvolution (lines with triangles). The enlargement of WN 1750 to 1550 cm^{-1} (right column) shows the region of C=O bands. Spectra were KM transformed and normalized to [0; 1] to allow for comparison.

Table 6 Number of analysed intact samples and DRIFT spectra ($n_{\text{total}} = 3049$) obtained from earthworm burrows, root channels, crack surfaces and mixed samples of the uncoated soil matrix. Mean values and standard deviations (in parentheses) of summed signal intensities of C–H and C=O bands

Site	Samples / n	Earthworm burrows			Root channels			Crack surfaces			Matrix (mixed samples)		
		Spectra / n	C–H / KM	C=O / KM	Spectra / n	C–H / KM	C=O / KM	Spectra / n	C–H / KM	C=O / KM	Spectra / n	C–H / KM	C=O / KM
haLV1(a)	4	319	0.76 (0.32)	39.92 (5.38)	–	–	–	70	0.33 (0.30)	31.62 (10.23)	3	0.017 (0.001)	16.88 (0.19)
haLV1(f)	4	107	1.29 (0.42)	37.53 (4.11)	108	2.05 (1.76)	24.68 (7.61)	32	0.28 (0.23)	29.14 (4.32)	3	0.023 (0.002)	16.29 (0.35)
haLV2	5	293	1.46 (0.95)	47.91 (6.45)	–	–	–	71	0.52 (0.24)	38.89 (7.35)	3	0.013 (0.001)	16.10 (0.14)
haLV3	3	86	0.19 (0.09)	18.72 (2.34)	33	0.08 (0.03)	18.76 (1.53)	61	0.05 (0.03)	18.13 (1.46)	3	0.013 (0.002)	14.35 (1.03)
abLV	9	242	0.46 (0.45)	25.05 (5.51)	157	0.47 (0.27)	28.31 (5.18)	121	0.16 (0.12)	25.41 (6.53)	3	0.025 (0.003)	14.82 (0.28)
haRG	8	159	2.33 (0.62)	33.56 (4.46)	91	2.84 (0.81)	33.79 (3.25)	85	2.03 (0.47)	34.08 (4.90)	3	0.653 (0.014)	19.51 (0.14)
vrST	5	130	2.47 (1.16)	46.95 (5.79)	81	3.16 (0.71)	56.14 (5.65)	82	2.13 (0.87)	40.68 (11.67)	3	0.677 (0.010)	20.42 (0.23)
vrCM	5	300	4.56 (1.19)	53.14 (7.77)	–	–	–	84	3.60 (1.15)	45.35 (11.09)	3	1.068 (0.047)	19.76 (0.32)
haCM	5	194	1.26 (0.92)	29.8 (11.71)	65	3.52 (0.96)	17.81 (4.74)	78	0.73 (0.63)	25.16 (4.15)	3	0.048 (0.010)	15.84 (1.11)

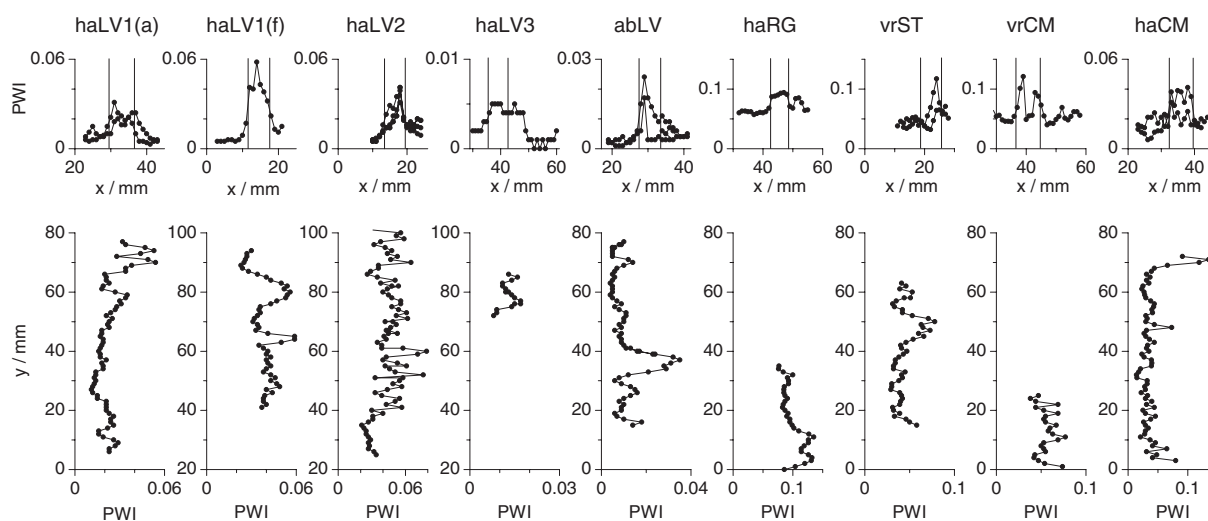


Figure 7 Potential wettability (PWI) values along (bottom row) and perpendicular to earthworm burrows (upper row) from five Luvisol (haHL, abLV), one Regosol (haRG; here root channel), one Stagnosol (vrST) and two Cambisol (vrCM, haCM) subsoil horizons. In the upper row the position of the earthworm burrows/root channel is denoted by the vertical lines.

be carried out because samples always have to be standardized (ground) and the millimetre-scaled spatial location of the SOM will be lost. In the subsoil, the application of ^{13}C NMR analyses is limited by small OC:Fe ratios. However, the alternative removal of Fe from the samples (by hydrochloric acid or hydrogen fluoride) may strongly alter the amount and composition of SOM (Dai & Johnson, 1999).

While the spectral region of the aliphatic C–H bands between $\text{WN } 3000$ and 2800 cm^{-1} was not influenced by absorption bands of other solid soil components (Figure 6), the position of the C=O bands between $\text{WN } 1750$ and 1500 cm^{-1} is influenced by interaction with metal cations (Piccolo & Stevenson, 1982) and the formation of hydrogen bonds and is complicating the peak identification. The broad band between 1750 and 1500 cm^{-1}

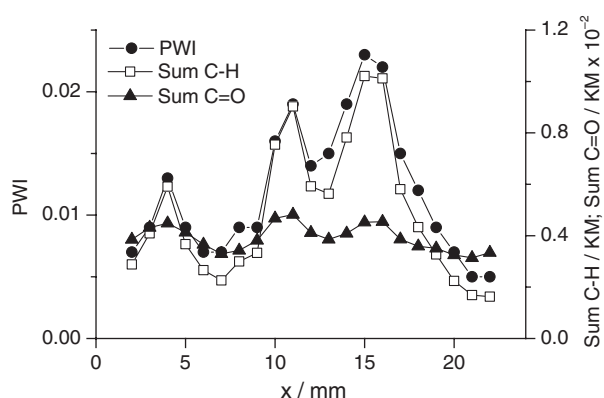


Figure 8 Single DRIFT measurement transect across an earthworm burrow of a haLV1(a) sample: potential wettability (PWI) (filled circles), summed C–H signal intensities (empty squares) and summed C=O signal intensities (filled triangles, multiplied by 10^{-2} for better comparison). Measurement points x from 10 to 17 mm are inside the burrow, points from 2 to 7 mm and 18 to 22 mm are located at neighbouring cracks.

can be ascribed to many individual C=O bands and absorption bands (aromatic C=C stretch), which may shift depending on the covalence of the metal-carboxylate complexes (Baes & Bloom, 1989).

The large spatial heterogeneity of the PWI along the root channel walls, reflected in extremely large standard deviations (Figure 9), can be attributed to the variable presence of decayed root material, in part strongly attached to the channel surfaces, and also to differing degrees of decay of this material (Rumpel & Kögel-Knabner, 2011).

OM composition at structural surface types

The differences between the PWI of uncoated matrix samples and intact structural surfaces such as biopores coated by clay-organic material (Figure 9) indicate the importance of OM input to the surfaces of subsoil structures via vertical transport processes such as clay illuviation, earthworm activity and root growth and to OM distribution in structured soils. Similar differences in the OM composition at intact structural surfaces to that of a mixed bulk sample have been reported before (Ellerbrock *et al.*, 2009). The comparison of the SOC contents (Table 2) with the PWI (Figure 9) confirms that OM composition and SOC content were not related; this is particularly true for coatings on structural surfaces (Ellerbrock *et al.*, 2005).

Except for the Vertic Stagnosol (vrST) where the PWI values were not significantly different between the surface types (Figure 9), the OM on the biopore walls was composed of relatively larger proportions of aliphatic groups than that at coated crack surfaces, emphasizing the influence of the soil biota on the properties of intact structural surfaces (Lee, 1985; Bundt *et al.*, 2001). The PWI values of earthworm burrow surfaces of haLV1(a) are consistent with those reported by Ellerbrock *et al.* (2009, who denoted PWI as an A:B ratio) for the same site. For the haLV2 samples, the

PWI values correspond to those previously obtained with DRIFT mapping (Leue *et al.*, 2010b; also denoted as A:B ratios).

Although the mean PWI values were significantly different for most surface types (Table 4), the ranges in PWI values were large (Figure 9), indicating a characteristic spatial heterogeneity of the OM composition on the different intact structural surfaces. The millimetre-scaled spatial heterogeneity of the OM composition along single transects (Figures 7, 8) corresponded to the results of a two-dimensional DRIFT mapping analysis of such surfaces (Leue *et al.*, 2010b). Such heterogeneity can be explained by the continuity and complex geometry of macropores (Köhne *et al.*, 2011), which control the process of clay illuviation (Lin, 2010) and can contribute to a spatial distribution of OM along PF path surfaces.

The heterogeneity of PWI values along the worm-burrow surfaces (Figures 7, 8) suggests that the walls contain regions of small and large wettability. Small-scale differences in OM composition along earthworm burrow walls can be attributed to their activity (Lee, 1985; Don *et al.*, 2008; Ellerbrock *et al.*, 2009). The broad range in PWI for earthworm burrows could possibly also be explained by a difference in age of the burrows (Leue *et al.*, 2010b). The fraction-dependent rates of OM decomposition for mucilage, earthworm casts and root residues could lead to a temporal heterogeneity in OM composition, which, in addition to the spatial variability, influences the potential wettability of the biopore surfaces.

Soil type-specific OM composition

Among the nine selected sites (Figure 9) the five Luvisols (haLV, abLV) form a relatively homogeneous group with respect to their OM composition. The PWI values indicate that the potential wettability of the surface types is larger for the Luvisol samples than for those of the other soil types. Here, the process of clay migration seems to affect both the composition and spatial distribution of OM on the structural surfaces of Bt horizons.

For the forested Haplic Regosol (haRG), the overall larger PWI value indicates that the OM composition of all structural surface types of the uncoated matrix sample is different from that of arable and forested Luvisols (Figure 9). The relatively larger proportions of aliphatic groups (C–H bands) in the OM of the biopore surfaces could be explained by incorporation of OM-rich topsoil material (forest litter) by bioturbation (Lee, 1985; Don *et al.*, 2008). Additionally, the pH of about 7.4 and the large content of Ca^{2+} ions (Kaiser *et al.*, 2008) indicate the possibility of strong interactions between OM and Ca^{2+} , which may lead to a stabilization of aliphatic-rich OM.

For the Stagnosol (vrST), the PWI of all structural surfaces and the uncoated matrix samples were similarly large and relatively heterogeneously distributed (Figures 9, 10). Such similarity may be explained by the influence of periodic water saturation ('stagnic' horizon), causing the mobilization of Fe- and Mn-sesquioxides (Hagedorn *et al.*, 2000) by redoximorphic processes and thus, the dissolution and movement of OM.

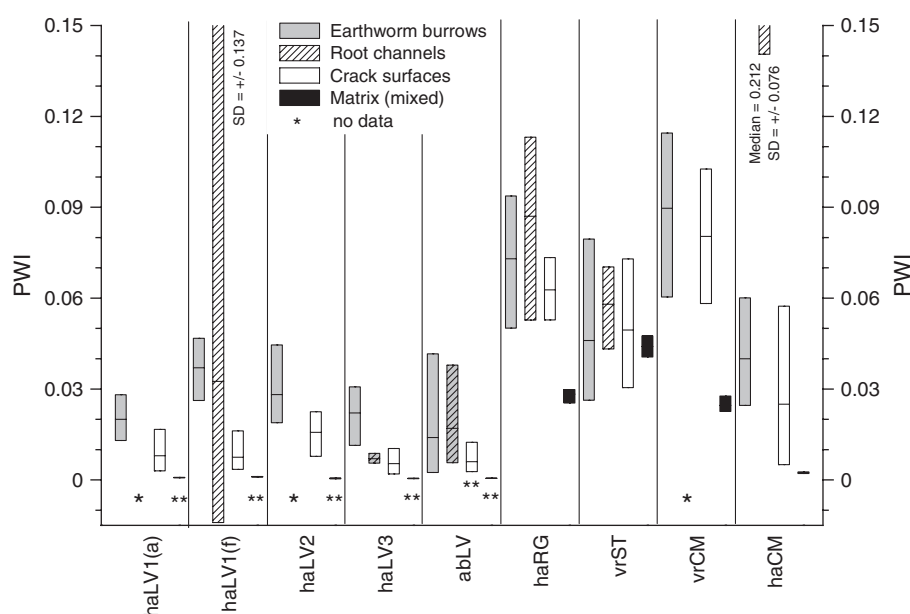


Figure 9 Potential wettability (PWI) of soil organic matter (SOM) of coatings at biopores (earthworm burrows, root channels), crack surfaces and mixed samples of the soil matrix (without coatings) from five Luvisol (haHL, abLV), one Regosol (haRG), one Stagnosol (vrST) and two Cambisol (vrCM, haCM) subsoil horizons, median and standard deviations (*, no data; **, C–H signal intensity of all mixed samples at some crack surface measurement points is less than the limit of detection of 0.02 KM units).

The large PWI levels of intact subsoil samples of the Vertic Cambisol (vrCM; Figure 9) can be explained by the interaction between OM and minerals (clay content about 60%; Table 2), which may stabilize the aliphatic OM groups (Lehmann *et al.*, 2007). However, incomplete removal of crop fine roots growing within biopores and cracks may have also caused similarly large PWI mean values and ranges at the earthworm burrow and crack surfaces of vrCM (Figure 9).

For the subsoil samples of the forested Haplic Cambisol (haCM) developed on limestone, the PWI values of both earthworm burrows and crack surfaces (Figure 9) were smaller than those of the Vertic Cambisol (vrCM), although the clay content was the largest of all subsoil samples (Table 2). In contrast, the root channel surfaces were dominated by OM with large proportions of aliphatic groups, which caused large PWI values and can be attributed to root residues from forest vegetation (Jandl *et al.*, 2012).

Influence of the parent material and land use

Even though the DRIFT spectra of the Luvisol subsoil samples were similar (Figure 6), the data indicated significant differences in the PWI values between loess-derived (haLV1(a), haLV1(f) and haLV2) and till-derived Luvisols (haLV2 and haLV3; Figure 9). Clay mineral absorption bands (WN 3700 to 3400 cm⁻¹) were found in all Luvisol spectra without any clear trend, suggesting that in the case of the Luvisol subsoils, the effect of clay illuviation was possibly more important for OM composition than differences in the parent materials.

The land-use effect on the OM composition at structural surfaces is most clearly seen by comparing data for loess-derived Luvisol samples from the arable (haLV1(a)) and forest sites (haLV1(f)), located about 200 m apart and showing a continuous land use for at least 200 years (Kaiser *et al.*, 2010). The SOC

contents in the Bt horizons were larger for the arable than for the forest site (Table 2), indicating a more intensive OM relocation under arable use. The composition and the greater variability of the OM input at the forest sites (Nierop, 1998) could explain the larger and more heterogeneously distributed PWI values at earthworm burrow walls (Figures 9, 10).

Conclusions

The distribution of the OM composition at intact structural surfaces of subsoil horizons was characterized by DRIFT spectroscopy. The ratio between the signal intensity of alkyl (C–H) and carbonyl (C=O) functional groups of the OM was used as a characteristic for OM composition in terms of the potential wettability (PWI) of the structural surfaces that could serve as PF paths in structured soil. The results suggest that the millimetre-scale spatial distribution of the OM composition of biopore and crack surfaces in subsoil horizons differs from the mean OM composition of the bulk soil depending on the:

- 1 structural surface: biopores had a smaller PWI (greater potential wettability) than crack surfaces;
- 2 soil type: structural surfaces of samples from Luvisols had smaller PWI values than samples from the Regosol, the Stagnosol and Cambisols;
- 3 parent material: structural surfaces of samples from loess-derived Luvisols had larger PWI values (less potential wettability) than those from till-derived Luvisols; and
- 4 land use: the PWI of structural surfaces from arable soils tended to be less than that from forest soil.

However, the complexity of soil formation limits a clear differentiation between possible effects of the activity of soil

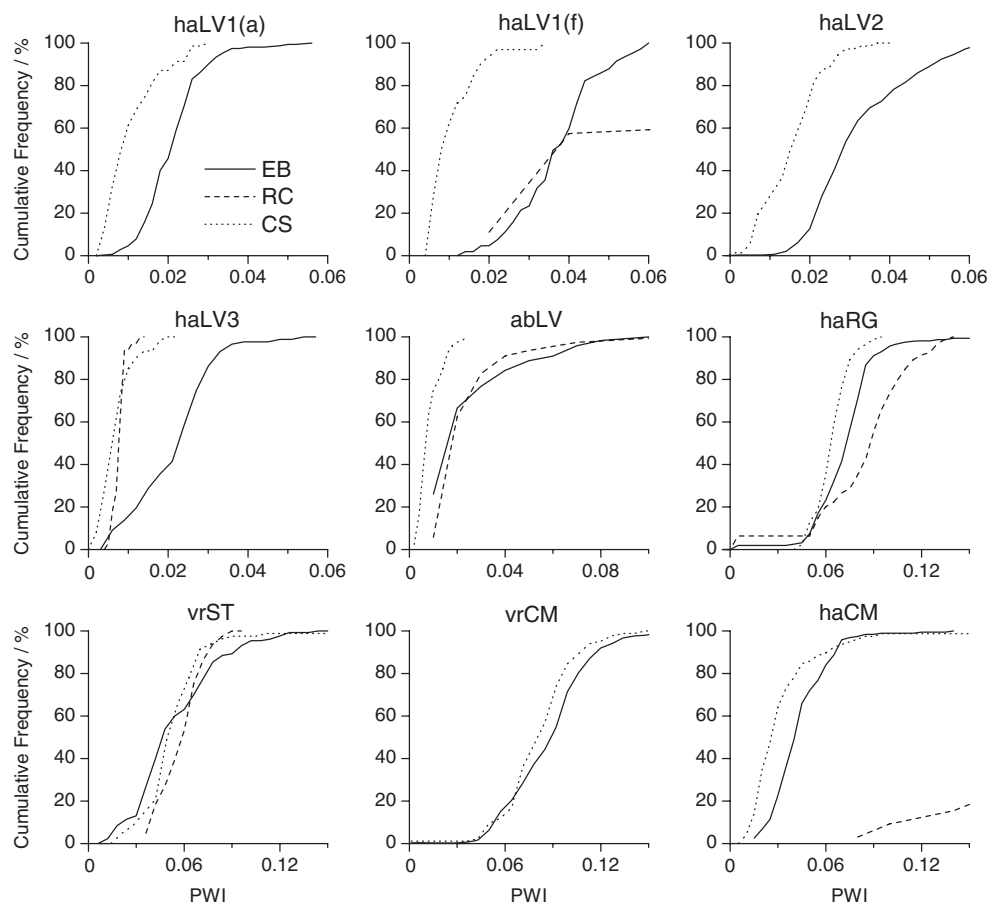


Figure 10 Cumulative distribution of potential wettability (PWI) values at three flow path types (EB, earthworm burrow; RC, root channel; CS, crack surface) from five Luvisol (haHL, abLV), one Regosol (haRG), one Stagnosol (vrST) and two Cambisol (vrCM, haCM) subsoil horizons.

biota, pedogenetic processes, parent material and land use on the OM composition at structural surfaces by means of DRIFT spectroscopy.

The millimetre-scale heterogeneity of OM composition suggests that in addition to the potential wettability, other coating properties of preferential flow path surfaces (cation exchange capacity, sorption properties, and mass transfer rates) may be spatially distributed in a characteristic way. To evaluate further such influences of small-scale properties on PF, it will be necessary to quantify SOM at intact structural surfaces *in situ*, to characterize the mineral components of these surfaces, and to analyse the geometric arrangement of crack and biopore structures in the soil volume.

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References

- Baes, A.U. & Bloom, P.R. 1989. Diffuse reflectance and transmission Fourier transform infrared (DRIFT) spectroscopy of humic and fulvic acids. *Soil Science Society of America Journal*, **53**, 695–700.
- Benesi, H. & Jones, A.C. 1959. An infrared study of the water-silica gel system. *Journal of Physical Chemistry*, **63**, 179–182.
- Bundt, M., Widmer, F., Pesaro, M., Zeyer, J. & Blaser, P. 2001. Preferential flow paths: biological 'hot spots' in soils. *Soil Biology & Biochemistry*, **33**, 729–738.
- Capriel, P., Beck, T., Borchert, H., Gronholz, J. & Zachmann, G. 1995. Hydrophobicity of the organic matter in arable soils. *Soil Biology & Biochemistry*, **27**, 1453–1458.
- Chabbi, A., Kögel-Knabner, I. & Rumpel, C. 2009. Stabilised carbon in subsoil horizons is located in spatially distinct parts of the soil profile. *Soil Biology & Biochemistry*, **41**, 256–261.

- Dai, K. & Johnson, C.E. 1999. Applicability of solid-state ^{13}C CP/MAS NMR analysis in Spodosols: chemical removal of magnetic materials. *Geoderma*, **93**, 289–310.
- Demyan, M.S., Rasche, F., Schulz, E., Breulmann, M., Müller, T. & Cadisch, G. 2012. Use of specific peaks obtained by diffuse reflectance Fourier transform mid-infrared spectroscopy to study the composition of organic matter in a Haplic Chernozem. *European Journal of Soil Science*, **63**, 189–199.
- Don, A., Steinberg, B., Schoening, I., Pritsch, K., Joschko, M., Gleixner, G. *et al.* 2008. Organic carbon sequestration in earthworm burrows. *Soil Biology & Biochemistry*, **40**, 1803–1812.
- Ellerbrock, R.H. & Gerke, H.H. 2004. Characterizing organic matter of soil aggregate coatings and biopores by Fourier transform infrared spectroscopy. *European Journal of Soil Science*, **55**, 219–228.
- Ellerbrock, R.H., Gerke, H.H., Bachmann, J. & Goebel, M.-O. 2005. Composition of organic matter fractions for explaining wettability of three forest soils. *Soil Science Society of America Journal*, **69**, 57–66.
- Ellerbrock, R.H., Gerke, H.H. & Böhm, C. 2009. In situ DRIFT characterization of organic matter compositions on soil structural surfaces. *Soil Science Society of America Journal*, **73**, 531–540.
- Fahrenhorst, A., Topp, E., Bowman, B.T. & Tomlin, A.D. 2000. Earthworms and the dissipation and distribution of atrazine in the soil profile. *Soil Biology & Biochemistry*, **32**, 23–33.
- Gerke, H.H. 2012. Macroscopic representation of the interface between flow domains in structured soil. *Vadose Zone Journal*, **11**. doi: 10.2136/vzj2011.0125.
- Hagedorn, F., Kaiser, K., Feyen, H. & Schleppi, P. 2000. Effects of redox conditions and flow processes on the mobility of dissolved organic carbon and nitrogen in a forest soil. *Journal of Environmental Quality*, **29**, 288–297.
- Jandl, G., Baum, C., Blumschein, A. & Leinweber, P. 2012. The impact of short rotation coppice on the concentrations of aliphatic soil lipids. *Plant & Soil*, **350**, 163–177.
- Jarvis, N.J. 2007. A review of non-equilibrium water flow and solute transport in soil macropores: principles, controlling factors and consequences for water quality. *European Journal of Soil Science*, **58**, 523–546.
- Kaiser, M., Ellerbrock, R.H. & Gerke, H.H. 2008. Cation exchange capacity and composition of soluble soil organic matter fractions. *Soil Science Society of America Journal*, **72**, 1278–1285.
- Kaiser, M., Wirth, S., Ellerbrock, R.H. & Sommer, M. 2010. Microbial respiration activities related to sequentially separated, particulate and water-soluble organic matter fractions from arable and forest topsoils. *Soil Biology & Biochemistry*, **42**, 418–428.
- Kauppinen, J.K., Moffatt, D.J., Mantsch, H.H. & Cameron, D.G. 1981. Fourier self-deconvolution: a method for resolving intrinsically overlapped bands. *Applied Spectroscopy*, **35**, 271–276.
- Kodesova, R., Jirku, V., Kodes, V., Mühlhanselova, M., Nikodem, A. & Zigova, A. 2011. Soil structure and soil hydraulic properties of Haplic Luvisol used as arable land and grassland. *Soil & Tillage Research*, **111**, 154–161.
- Köhne, J.M., Köhne, S. & Šimůnek, J. 2009. A review of model applications for structured soils: b) Pesticide transport. *Journal of Contaminant Hydrology*, **104**, 36–60.
- Köhne, J.M., Schlüter, S. & Vogel, H.-J. 2011. Predicting solute transport in structured soil using pore network models. *Vadose Zone Journal*, **10**, 1082–1096.
- Kortüm, G., Braun, W. & Herzog, G. 1963. Principles and techniques of Diffuse-Reflectance Spectroscopy. *Angewandte Chemie (Applied Chemistry, International Edition)*, **2**, 333–341.
- Kubelka, P. 1948. New contributions to the optics of intensely light-scattering materials. Part I. *Journal of the Optical Society of America*, **38**, 448–457.
- Lee, K.E. 1985. *Earthworms – Their Ecology and Relationships with Soils and Land Use*. Academic Press, New York.
- Lehmann, J., Kinyangi, J. & Solomon, D. 2007. Organic matter stabilization in soil microaggregates: implications from spatial heterogeneity of organic carbon contents and carbon forms. *Biogeochemistry*, **85**, 45–57.
- Leue, M., Ellerbrock, R.H. & Gerke, H.H. 2010a. Impact of soil microstructure geometry on DRIFT spectra: comparisons with Beam Trace Modeling. *Soil Science Society of America Journal*, **74**, 1976–1986.
- Leue, M., Ellerbrock, R.H. & Gerke, H.H. 2010b. DRIFT-mapping of organic matter composition at intact soil aggregate surfaces. *Vadose Zone Journal*, **9**, 317–324.
- Leue, M., Gerke, H.H. & Ellerbrock, R.H. 2011. Correcting microtopography effects on DRIFT mapping signals of organic matter at intact soil aggregate surfaces. *Soil Science Society of America Journal*, **75**, 1626–1639.
- Lin, H. 2010. Linking principles of soil formation and flow regimes. *Journal of Hydrology*, **393**, 3–19.
- Moradi, A.B., Carminati, A., Lamparter, A., Woche, S.K., Bachmann, J., Vetterlein, D. *et al.* 2012. Is the rhizosphere temporarily water repellent? *Vadose Zone Journal*, **11**. doi: 10.2136/vzj2011.0120.
- Nierop, K.G.J. 1998. Origin of aliphatic compounds in a forest soil. *Organic Geochemistry*, **29**, 1009–1016.
- Péré, E., Cardy, H., Cairon, O., Simon, M. & Lacombe, S. 2001. Quantitative assessment of organic compounds adsorbed on silica gel by FTIR and UV-VIS spectroscopies: the contribution of diffuse reflectance spectroscopy. *Vibrational Spectroscopy*, **25**, 163–175.
- Piccolo, A. & Stevenson, F.J. 1982. Infrared spectra of Cu^{2+} Pb^{2+} and Ca^{2+} complexes of soil humic substances. *Geoderma*, **27**, 195–208.
- Rumpel, C. & Kögel-Knabner, I. 2011. Deep soil organic matter – a key but poorly understood component of terrestrial C cycle. *Plant & Soil*, **338**, 143–158.
- Senesi, N., D'Orazio, V. & Ricca, G. 2003. Humic acids in the first generation of Eurosoils. *Geoderma*, **116**, 325–344.
- WRB 2006. *World Reference Base for Soil Resources 2006*. World Soil Resources Report No 103, FAO, Rome.